Solid-state electron spin lifetime limited by phononic vacuum modes

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(Dated: June 30, 2017)

Longitudinal relaxation is the process by which an excited spin ensemble decays into its thermal equilibrium with the environment. In solid-state spin systems relaxation into the phonon bath usually dominates over the coupling to the electromagnetic vacuum. In the quantum limit the spin lifetime is determined by phononic vacuum fluctuations. However, this limit was not observed in previous studies due to thermal phonon contributions or phonon-bottleneck processes. Here we use a dispersive detection scheme based on cavity quantum electrodynamics (cQED) to observe this quantum limit of spin relaxation of the negatively charged nitrogen vacancy (NV^-) centre in diamond. Diamond possesses high thermal conductivity even at low temperatures, which eliminates phonon-bottleneck processes. We observe exceptionally long longitudinal relaxation times T_1 of up to 8 h. To understand the fundamental mechanism of spin-phonon coupling in this system we develop a theoretical model and calculate the relaxation time *ab initio*. The calculations confirm that the low phononic density of states at the NV^- transition frequency enables the spin polarization to survive over macroscopic timescales.

arXiv:1706.09798v1 [cond-mat.mes-hall] 29 Jun 2017

The longitudinal relaxation of electron spins trapped in insulating solids has extensively been studied in the past - both theoretically [1–5] and in experiment [6–13]. A theoretical treatment of spin-lattice interaction results in two-phonon Raman [14] and Orbach [4] processes, dominating at higher temperatures and a direct single phonon process in the low temperature regime. If all thermal phonons are frozen out the ultimate limit of spin lifetime is given by the coupling of the spin system to the phononic vacuum modes. The observation of this limit can be obscured by a phonon bottleneck [15–17], where excited spins decay and are re-excited if the local phonon excitations are not transferred out of the system sufficiently rapid.

A suitable spin system to observe the coupling to phononic vacuum modes is the NV^- centre in diamond. Diamond has a high thermal conductivity even at low temperatures [18] which suppresses bottleneck processes. The ground state spin triplet of the NV^- centre is split due to dipolar spin-spin interaction by a zero-field splitting of $\omega_s/2\pi = D/h = 2.88\,\mathrm{GHz}$ at low temperatures [19, 20] into two upper ($m_s = \pm 1$) and a lower ($m_s = 0$) level [19, 21]. With a base temperature of $\sim 25\,\mathrm{mK}$, our experimental apparatus can reach temperatures at which the thermal phonon occupation ($\bar{n} \sim e^{-5.5} \approx 4 \times 10^{-4}$) at the spin transition frequency is mostly suppressed ($138\,\mathrm{mK} \mathrel{\widehat{=}} 2.88\,\mathrm{GHz}$).

To investigate the spin-lattice interaction in this system we employ a cQED [22–25] protocol where we couple the spins to the oscillating electromagnetic field of a resonator to non-perturbatively read out the spin state as illustrated in Fig. 1a. The interaction of a spin-ensemble with the resonator mode is described by a modified Tavis-Cummings Hamiltonian [26] for a three level system with two degenerate excited states (see Methods). In the dispersive limit the resonator frequency $\omega_c/2\pi$ and the spin transition frequency $\omega_s/2\pi$ have a detuning that is much larger than the collective coupling g_N of the spin ensemble to the resonator mode ($\delta = \omega_c - \omega_s \gg g_N$). In our experiments we maintain a detuning $\delta/2\pi > 100\,\mathrm{MHz}$, corresponding to a negligible enhancement of the single spin spontaneous emission rate due to the Purcell effect [27, 28] of $\Gamma_p \approx 10^{-14}\,\mathrm{s}^{-1}$. The dispersive coupling results in a state dependent shift χ of the resonator frequency given by

$$\chi(t,T) = \frac{N g_0^2}{\delta}(2 - 3\,\langle S_z^2(t,T)\rangle). \tag{1}$$